YAN ZHANG

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EDUCATION

Doctor of Philosophy, Statistical Science September 2023 – Present
University of Toronto

- Supervisors: Stanislav Volgushev, Lei Sun

Master of Philosophy, Statistics and Actuarial Science September 2020 – September 2022
The University of Hong Kong

- Thesis: Monte Carlo methods for variance reduction and log-rank test

Bachelor of Science, Information and Computing Science September 2016 – July 2020
Harbin Institute of Technology

- Major GPA: 94.68/100 (3.9/4.0)

PUBLICATIONS & PREPRINTS

Preprints

- **Zhang, Y.**. (2026). Fast computation and theoretical guarantees for the NPMLE in exponential family mixtures. *arXiv:2604.19725*.
- **Zhang, Y.**, & Volgushev, S. (2026). Adaptivity of the NPMLE to finitely discrete mixing distributions in Gaussian/Poisson mixtures. *arXiv:2604.12087*.
- **Zhang, Y.**, & Volgushev, S. (2025). On a surprising behavior of the likelihood ratio test in non-parametric mixture models. *arXiv:2509.05610*.

Peer-Reviewed Articles

- Gu, J., **Zhang, Y.**, & Yin, G. (2023). Bayesian log-rank test. *The American Statistician*, 77(3), 292–300.
- **Zhang, Y.**, Lv, J., & Zou, X. (2020). Dynamics of stochastic single-species models. *Mathematical Methods in the Applied Sciences*, 43(15), 8728–8735.
- Lv, J., **Zhang, Y.**, & Zou, X. (2019). Recurrence and strong stochastic persistence of a stochastic single-species model. *Applied Mathematics Letters*, 89, 64–69.

RESEARCH & WORK EXPERIENCE

Research Assistant July 2022 – December 2022
University of Toronto

- Developed data-driven methods to infer graphical structures of multivariate extremes.
- Identified and resolved issues in the R package `graphicalExtremes`, improving stability and usability.

Data Analyst January 2022 – April 2022
Geological Engineering Survey Institute, Shanxi

- Built a recurrent neural network model to evaluate the impact of mining activities (e.g., karst water extraction, coal mining) on Liulin Spring's discharge patterns, enabling more accurate hydrological forecasting.

Research Assistant September 2020 – September 2021
The University of Hong Kong

- Proposed and analyzed a Monte Carlo framework integrating variance reduction techniques (e.g., importance sampling, control variates), achieving faster convergence compared to standard Monte Carlo methods.

Research Assistant September 2019 – April 2020
École Centrale de Nantes

- Designed an end-to-end machine learning pipeline for gesture recognition from electromyographic (EMG) signals, improving classification accuracy for biomechanical applications.

Network Operation Assistant July 2019 – September 2019
China Mobile Communications Group Co., Ltd.

- Assisted in designing C++ modules for satellite network resource allocation, improving system performance.
- Contributed to the development of a recommendation system using low-rank matrix factorization and deep neural networks, enhancing predictive accuracy.

TEACHING EXPERIENCE

Teaching Assistant September 2022 – Present
University of Toronto

- Led tutorials for undergraduate courses (STA220H1F, STA255H1S, STA237H1F, STA447H1S, STA302H1F, STA414H1S), provided academic support, and facilitated problem-solving sessions.

Teaching Assistant January 2021 – April 2022
The University of Hong Kong

- Led tutorials for a master-level course (`STAT8019 Marketing Analytics`) and a PhD-level course (`STAT6011 Computational Statistics`), guiding students through advanced statistical concepts.

AWARDS

Canada Graduate Research Scholarship – Doctoral University of Toronto
120,000 CAD *2026 – 2029*

Ontario Graduate Scholarship – International University of Toronto
15,000 CAD *2025 – 2026*

Arts & Science International Graduate Scholarship University of Toronto
5,000 CAD *2023 – 2024*

People's Scholarship, First Prize Harbin Institute of Technology
1,000 CNY (awarded twice) *2018, 2019*

People's Scholarship, Second Prize Harbin Institute of Technology
500 CNY (awarded three times) *2017, 2018, 2019*

Meritorious Winner, Mathematical Contest in Modeling 2019

First Prize, Contemporary Undergraduate Mathematical Contest in Modeling 2017